

Census Income Intelligence Platform

Project Report for Retail Client

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Dataset	1994-95 US Census Bureau - 199,523 individuals
Objective	Income classification and customer segmentation for retail marketing
Models	XGBoost (AUC-ROC: 0.9546) + KMeans Clustering (k=5, Silhouette: 0.2717)
Key Result	10.5x ROI improvement over random targeting at threshold 0.5

1. Business Problem

The retail client approached this engagement with a clear objective: improve the efficiency of marketing campaigns by identifying individuals most likely to earn over \$50,000 per year. Contacting an entire population without prioritization is both expensive and inefficient. A predictive model allows the client to focus spending on high-income individuals, reducing wasted budget and improving campaign returns.

A second requirement was to understand the natural groupings within the target population. Different customer types have different needs and respond to different messages. A segmentation model gives the client the structure to run tailored campaigns rather than one broad, untargeted effort.

Two deliverables: (1) A classifier that predicts income group for any individual using 40 demographic and employment variables, and (2) a segmentation model that groups the population into five actionable marketing segments.

2. Data Exploration

2.1 Dataset Overview

Source	1994-95 US Census Bureau (PUMS)
Records	199,523 individuals
Features	40 demographic and employment variables
Target	Binary - income over \$50K or under \$50K
Imbalance	94.0% under \$50K / 6.0% over \$50K

2.2 Class Imbalance

The first thing that stood out when I opened the dataset was the severe class imbalance. 94% of individuals earn under \$50K - only 6% earn over it. This is not unusual for income datasets, but it has a direct impact on every modeling decision downstream.

A model that blindly predicts "under \$50K" for everyone would score 94% accuracy. That sounds impressive but is completely worthless for the client's goal. Because of this, accuracy was never used as a metric in this project. AUC-ROC and F1 score were used throughout.

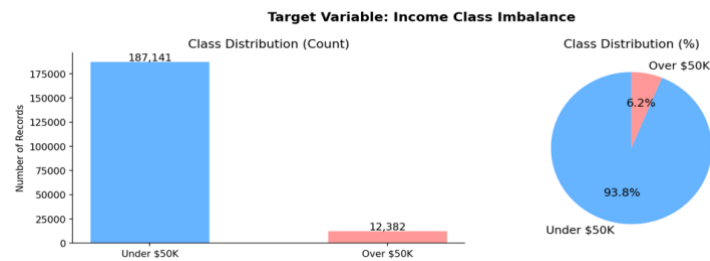


Figure 1: Target variable distribution - 94% under \$50K, 6% over \$50K

2.3 Key Findings from EDA

Several patterns emerged from the exploratory analysis that directly shaped the modeling approach:

- **Education is the strongest categorical predictor.** Professional and doctorate degree holders earn over \$50K more than 50% of the time. High school graduates sit at just 3.9%. The jump from high school to a bachelor's degree alone is from 3.9% to 19.7%.
- **Investment income is a disproportionately strong signal** despite being sparse in the data. Only 2.7% of low earners have any capital gains, compared to 19.5% of high earners - a 7x difference. Dividends show an even wider gap: 8.5% vs 42.9%.
- **Age peaks between 35 and 55.** The 45-54 age group has the highest income rate at 16.3%. Under 25 is essentially zero (0.2%), and the rate drops sharply after retirement age.
- **Self-employed incorporated workers** earn over \$50K at 34.7% - more than five times the overall average of 6.2%. This likely reflects established, profitable business ownership.
- **Four migration columns** were missing for exactly 49.97% of records - likely because the survey only asked these questions to a subset of respondents. These were dropped entirely rather than imputed.

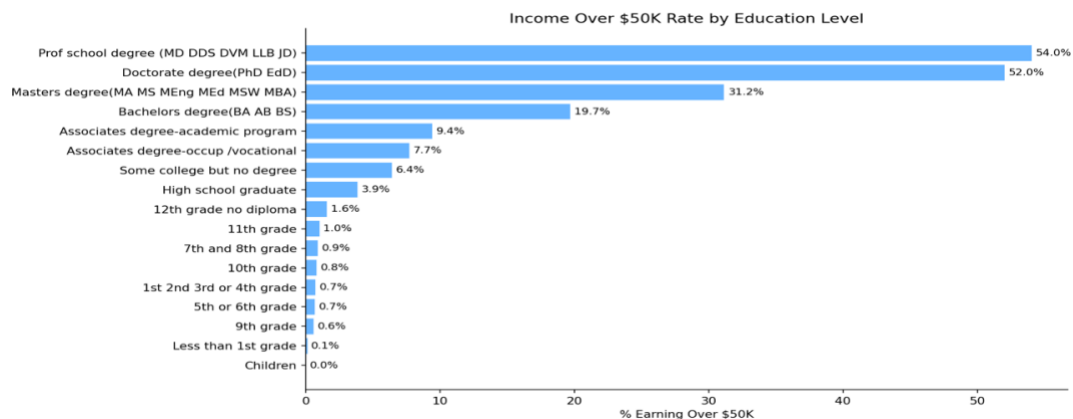


Figure 2: Income rate by education level - clear relationship from children through to professional degree

3. Classification Model

3.1 Pre-processing

The following steps were applied before model training:

- Dropped 4 migration columns with ~50% missing values
- Imputed remaining missing values with the mode for categorical features and the median for numeric features
- Dropped weight and year columns - weight is a survey correction factor, year has only two values (94/95), and adds no predictive signal
- One-hot encoded all categorical features. Feature names were sanitised to remove special characters for XGBoost compatibility
- StandardScaler applied to all numeric features
- 80/20 stratified train/test split to preserve the 94/6 class ratio in both sets

3.2 Model Comparison

Three models were trained and evaluated. All used stratified splits and class weighting to address the imbalance.

Model	AUC-ROC	F1 (>\$50K)	Precision	Recall	Selected
Logistic Regression	0.9460	0.4358	0.6821	0.3191	
Random Forest	0.9402	0.4299	0.7014	0.3089	
XGBoost	0.9546	0.4911	0.7198	0.3721	YES

XGBoost was selected as the final model based on its superior AUC-ROC and F1 score. Notably, Logistic Regression - the simplest model - performed competitively at 0.946, suggesting the data contains strong linear relationships that even a basic model can exploit. Random Forest slightly underperformed both, which was unexpected given its complexity.

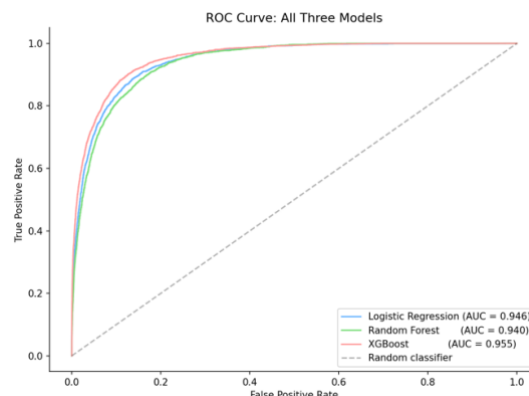


Figure 3: ROC curves for all three models - XGBoost (pink) leads, but all three perform well above the random baseline

3.3 Modeling Decisions

Decision	Detail
Class imbalance handling	scale_pos_weight=15 for XGBoost (ratio of negatives to positives). class_weight="balanced" for Logistic Regression and Random Forest.
Hyperparameter tuning	Two rounds of RandomizedSearchCV performed. Both rounds confirmed default XGBoost parameters were optimal - no further tuning required.
Cross-validation	5-fold stratified CV confirmed stability. Mean AUC: 0.9519, Std: 0.0018. The low standard deviation indicates consistent generalisation across folds.
Probability calibration	Platt scaling applied to correct XGBoost probability overconfidence on imbalanced data. AUC remained unchanged at 0.9546 after calibration. Calibrated probabilities are essential for the threshold-based decisions the client will make.
Explainability	SHAP analysis performed. Top features by mean absolute impact: age (1.85), weeks worked (1.03), tax filer Nonfiler (0.58), sex Male (0.40), dividends from stocks (0.28).
Fairness check	AUC verified across sex and education groups. Female AUC: 0.9423, Male AUC: 0.9472. The gap is small and within acceptable range.

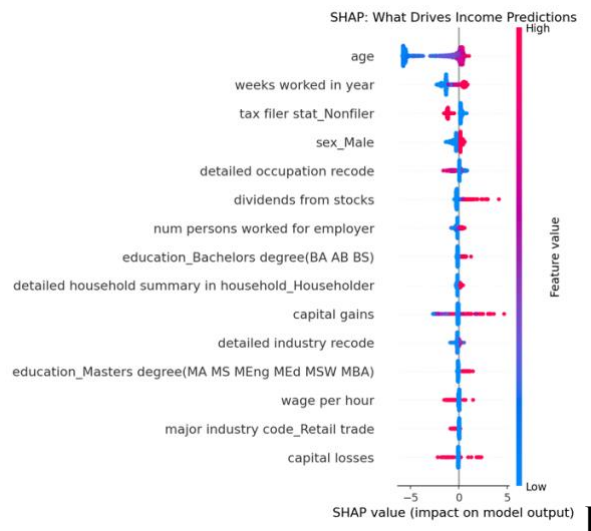


Figure 4: SHAP summary - age and weeks worked drive predictions most strongly, investment income consistently pushes predictions towards high income

3.4 Business Impact

The AUC number alone does not tell the client how much money this saves. The more useful framing is: how many more high-income individuals does this model find compared to random targeting?

ROI at Threshold 0.5: In a population of 10,000 people, random targeting finds approximately 620 high earners (the 6.2% base rate). Using this model at threshold 0.5, the same effort identifies approximately 6,527 high earners - a **10.5x improvement**. The client can achieve the same results with a fraction of the marketing budget or find far more high-income customers for the same spend.

The model also supports flexible threshold selection depending on campaign strategy:

Threshold	Strategy	Precision	Recall	Use Case
0.3	Broad Reach	~45%	~85%	Maximum coverage. Good for awareness campaigns.
0.5	Balanced	~62%	~65%	Default - best overall ROI. Recommended starting point.
0.7	Precision	~94%	~35%	Highly targeted. Best for premium high-spend campaigns.

4. Customer Segmentation

4.1 Approach

K-Means clustering was applied to discover natural marketing groups in the population. Unlike the classification model, the income label was deliberately excluded from the clustering process. The goal was to find groups based on who people are and how they live, then examine how income distributes across those groups as a validation step afterward.

4.2 Feature Selection

14 features were selected from the original 40. Every dropped column has a documented reason:

- **Sparse features dropped:** wage per hour (94.3% zeros), capital gains raw (96.3% zeros), capital losses (98.0% zeros), dividends raw (89.4% zeros). Capital gains and dividends were replaced with binary flags, has any vs has none, which captures the signal more cleanly without the distortion from extreme outliers.
- **High-cardinality features dropped:** detailed occupation recode (47 unique values), detailed industry recode (52 unique values), country-of-birth columns (42 unique values). These are numeric lookup codes where a higher number does not mean more of anything, meaningless for K-Means distance calculations.
- **Race is excluded on ethical grounds** to prevent the algorithm from creating ethnicity-defined marketing segments. This was a deliberate business decision made before any modeling began.

- **Redundant features dropped:** Hispanic origin overlaps with citizenship, detailed household summary overlaps with marital status, and enrollment in education overlaps with education level.

4.3 Preprocessing for Clustering

- **Education** was ordinal encoded (0 to 16) to preserve natural order from Children through to Professional degree. Standard label encoding would assign arbitrary numbers making the distances between education levels meaningless.
- **Weeks worked** was binned into 4 groups: 0=not working, 1=part year, 2=most of year, 3=full year. The raw column had 48% zeros creating a bimodal distribution that would distort K-Means centroids.
- **StandardScaler applied.** Without scaling, age (range 0-90) would completely dominate binary features (range 0-1) in the distance calculations.

4.4 Choosing k=5

The elbow method and silhouette scores were computed for k=2 through k=10. The elbow curve flattened around k=5. Silhouette scores continued improving slightly up to k=9 (0.2877), but the gain from k=5 (0.2717) to k=9 was only 0.016, not significant enough to justify nearly double the number of segments. A retail marketing team can realistically act on 5 distinct segments; 8 or 9 becomes operationally unmanageable. k=5 was selected.

4.5 The Five Segments

Segment	Size	Income >\$50K	Avg Age	Key Characteristics
0 - Working Adults (Lower Edu)	19.7%	9.3%	39	Female, clerical occupation, high school education, married
1 - Children & Dependents	27.6%	0.0%	8	Not working, never married, tax nonfiler
2 - Working Adults (Some College)	25.6%	11.4%	38	Male, professional specialty, some college, married
3 - High Earner Investors	3.7%	32.7%	48	100% have capital gains, mid-career, married
4 - Retired / Elderly	23.4%	1.1%	55	Female, not working, tax nonfiler, older age profile

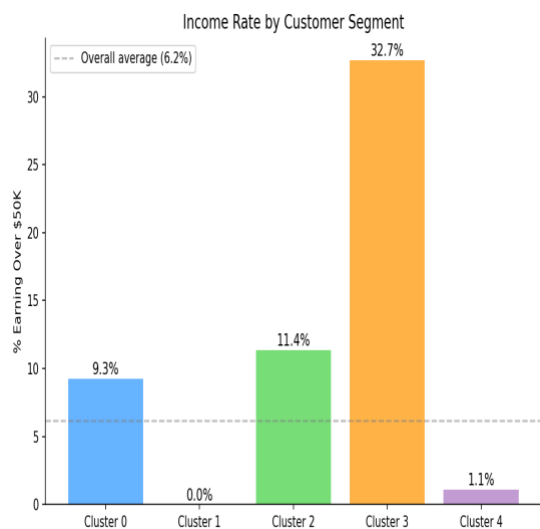


Figure 5: Income rate by segment - Segment 3 earns over \$50K at 5x the population average



Figure 6: customer segments in 2D using PCA - Segment 1(Children, pink) clearly separates on the left

Segment 3 - High Earner Investors: Despite representing only 3.7% of the population, this segment earns over \$50K at 32.7% - more than five times the overall average of 6.2%. Every single person in this segment has investment income. These are mid-career professionals aged 35-65 with established investment portfolios. Premium marketing spend is most justified here.

5. Marketing Recommendations

5.1 Segment Prioritization

Segment	Priority	Products	Channels	Message
Seg 3 - High Earner Investors	HIGH VALUE	Premium goods, financial services, investment products	Direct mail, premium digital	Quality, exclusivity, wealth management
Seg 2 - Working Adults (College)	ABOVE AVERAGE	Mid-range premium, home improvement, family products	Email, social media	Value, reliability, aspiration
Seg 0 - Working Adults (HS)	ABOVE AVERAGE	Mid-range products, family goods	Email, social media	Savings, practical value
Seg 4 - Retired / Elderly	LOW	Minimal spend recommended		Not recommended
Seg 1 -Children & Dependents	NONE	No spend		Exclude entirely

5.2 Using Both Models Together

The classification and segmentation models are most powerful when used in combination. The recommended workflow is:

1. Use the **segmentation model** to identify which segment a person belongs to. This determines which campaign type and message to assign.
2. Use the **classification model** to score their probability of earning over \$50K. This determines how much budget to allocate to this individual.
3. Apply the **threshold appropriate to the campaign type** - 0.3 for broad reach, 0.5 for standard campaigns, 0.7 for premium spend targeting Segment 3.

The interactive Streamlit application delivered with this project implements exactly this workflow. The client can enter an individual's details and receive both a probability score and a plain-English recommendation immediately.

6. Observations, Questions, and Future Work

6.1 Things Worth Raising

- **The dataset is from 1994-95.** Income thresholds, education patterns, and employment structures have changed significantly in 30 years. The \$50K threshold from 1994 is equivalent to roughly \$100K+ today when adjusted for inflation. I would want to understand whether the model will be applied to a similarly dated population or to current data before recommending deployment.
- **The gender pay gap is baked into the model.** Sex (Male) is the 4th most important SHAP feature because men earned more than women in the 1994 data. Using this model for current targeting may perpetuate that historical bias, worth a conversation with the client before deployment.
- **Segments 0 and 2 are very similar** in their numeric profiles. Both are working-age married adults with above-average income rates. The main differentiator is occupation and sex. The client may find it simpler to treat them as a single "working adult" segment depending on campaign complexity.

6.2 Questions for the Client

- Will the model be applied to a current population or a similarly dated census dataset? This affects whether retraining is necessary before deployment.

- What is the cost per contact and the average revenue per converted high-income customer?
With these numbers, we can calculate the exact financial ROI at each threshold rather than the relative 10.5x figure.
- Is Segment 3 large enough in the client's actual target market to justify a dedicated premium campaign, given it is only 3.7% of the population?

6.3 Future Improvements

- **Retrain on newer data.** The 1994 Census data is a proxy. A model trained on current income and employment data would be significantly more accurate for modern targeting.
- **Add behavioural features.** Current spending behaviour, purchase history, or online activity would dramatically improve both models if the client can provide this data.
- **Individual SHAP explanations in the app.** Currently the app shows a probability score. Adding a per-prediction explanation, "this person scored high because of age and investment income", would help the client understand and trust the model on a case-by-case basis.
- **A/B test the threshold.** The 10.5x ROI is theoretical based on test set performance. Running a real campaign with model predictions versus a random control sample would validate the actual lift in the field.

7. References

Dataset	US Census Bureau PUMS 1994-95. UCI Machine Learning Repository. https://archive.ics.uci.edu/ml/datasets/Census-Income+(KDD)
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